1. Application Structure

The application works by the user creating a competition file with the extension of '.hya' through the interface of the application. By doing so the users can view this newly created file through the file explorer and may open it again through the relevant options over the interface of the application.

2. The Components

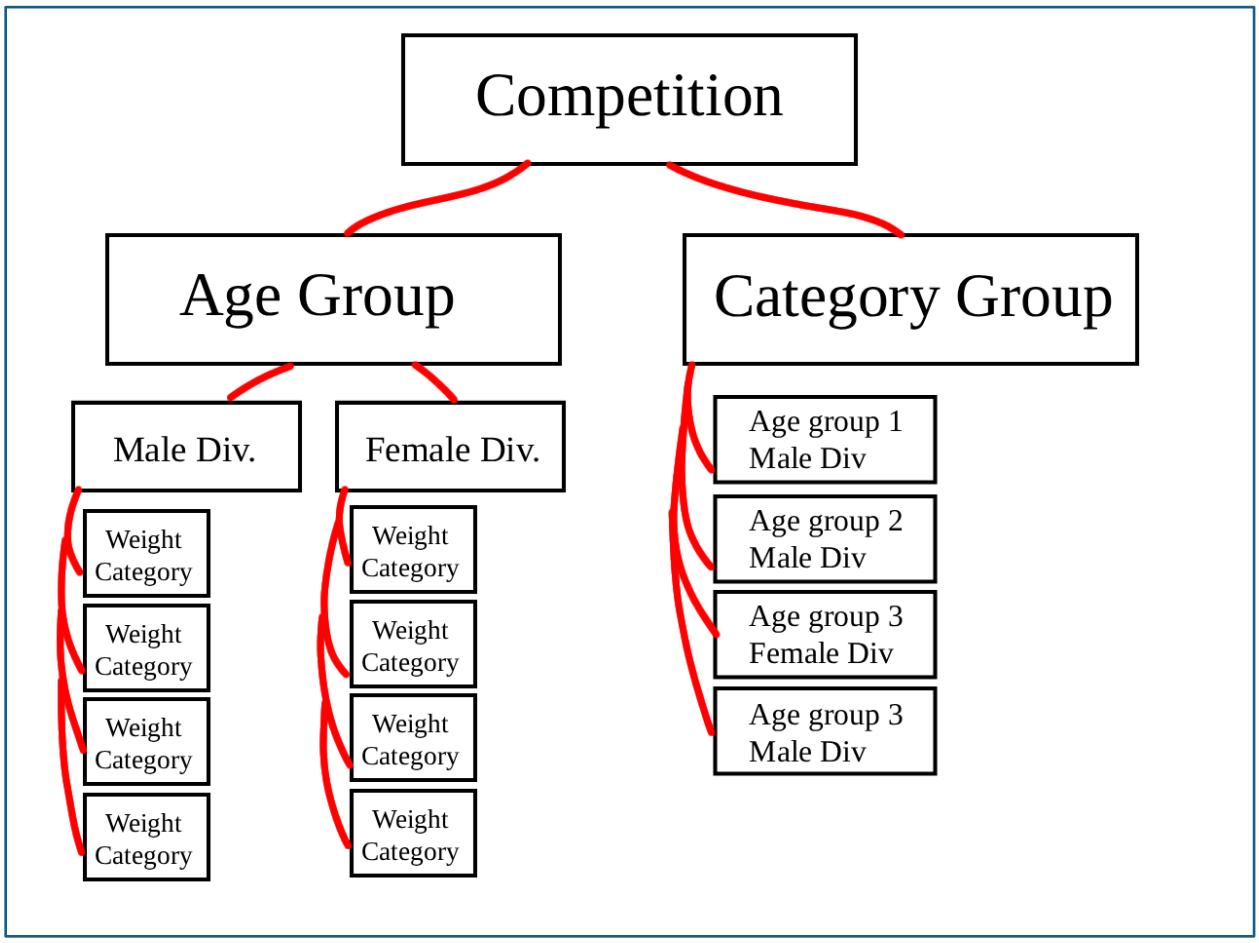


Figure 1.1. Component Diagram.

In *Figure 1.1* the diagram of the components through *2.1* and *2.5* have been depicted.

2.1. Competition

The biggest component of all is the Competition. It compasses all of the other components and is saved to a competition file with the previously mentioned extension. Upon opening the application, the users are supposed to create one of their own.

2.2. Age Group

Each age group has 2 divisions, Male division and the Female Division.

2.3. Category Group

Each category group can have a number of Divisions. The matches of each division are numbered respective to their position on the Category Group Division Manager menu.

2.4. Division

It has multiple weight categories and refers to one of the gender of an Age Group.

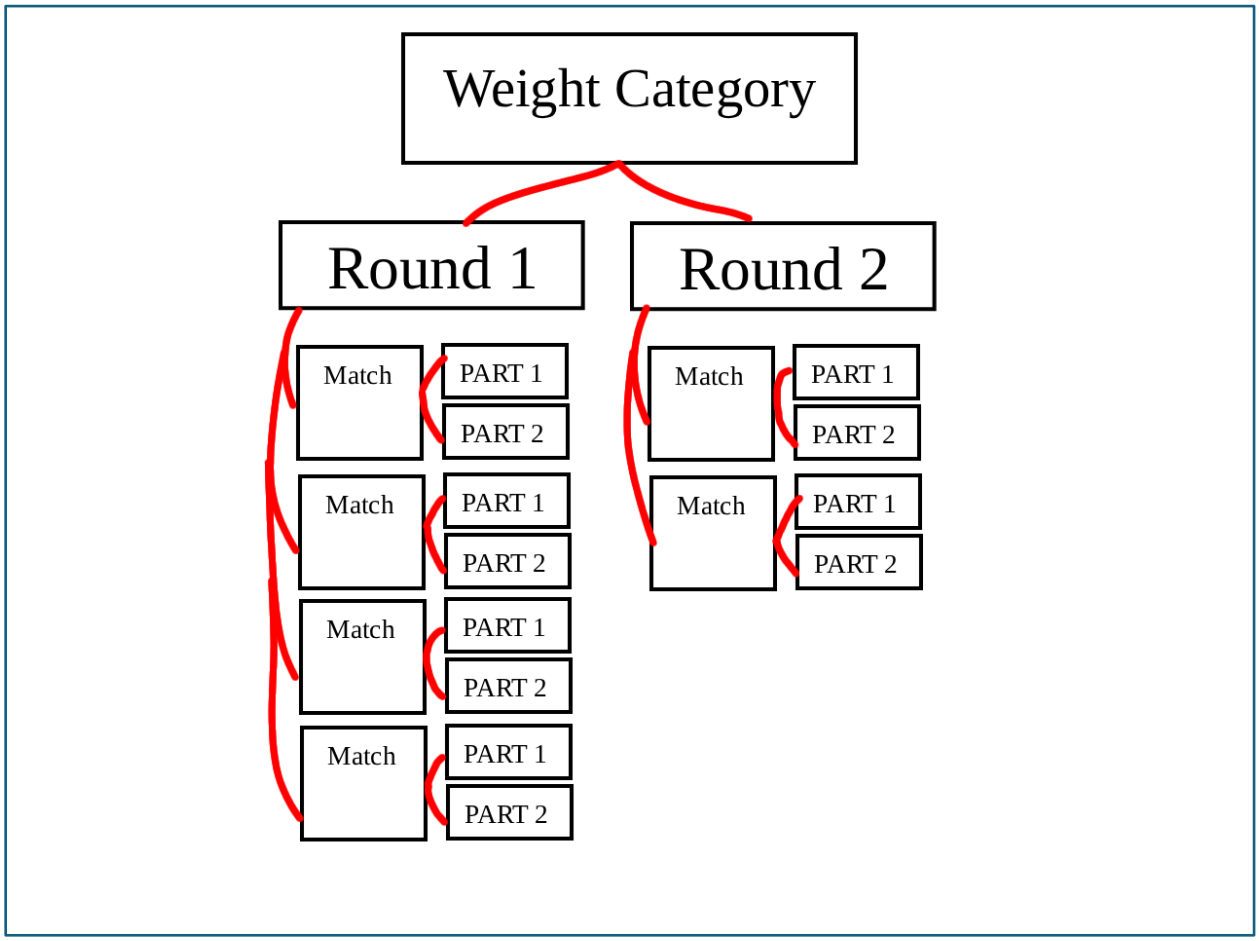


Figure 2.2. Weight Category Belligerents.

Over Figure 2.2 the belligerent of component *2.5* - components through *2.5* and *2.8* - have been depicted.

2.5. Weight Category

The weight category has multiple rounds. It refers to a certain weight range, gender and age group.

2.6. Round

A round is the segments you see in brackets. Each round has multiple matches, and they get progressively smaller as the brackets proceed.

2.7. Match

A match has 2 participants and a winner participant.

2.8. Participant

It can be considered as the core component of the entire process. It contains the core information about the Name, Date of Birth, Age, Gender etc.

3. The Structure

The biggest component of the structure is the Main Window which refers to the window that is initially opened upon launching the application. There are 2 main components here in this window.

3.1.1. Competition Viewer

In this component the user may view previously opened competition files and navigate through them.

3.1.2. Tab View

In this component there are 3 working tabs which the user can navigate through using the header.

3.1.2.1. Overview

In this tab users may change the properties of the application, add/remove and edit participants.

3.1.2.2. Groups

In this tab users may add/remove and modify 'Category Groups' which have individual 'divisions' of their own that can be added/removed by using this tab. By doing so they may view the entirety of the matches of the divisions added to a singular category.

3.1.2.3. Kumite

In this tab users may view several Age Groups, their divisions and their weight categories. Additionally, they may view the bracket of a singular selected weight category.

4. Features

4.1. Creating and Opening an Application/Competition File

Upon opening the application, navigate to the 'File' section located over the top-left corner. Here you may select; New File, Open File and Save File. Upon creating or opening a file a new entry will be added to the Competitions section just below the File section. You may click on the label of the added Competition to view it over the left Tab View panel.

4.2. Changing the Properties of the Competition

On the header of the Tab View panel click the tab labeled 'Overview'. In the Overview tab on the top, you can see a properties section. In this section you can modify them by clicking on the Text Boxes and pressing enter when done. However, for the Date property you must enter it in 'mm/dd/yyyy' format or else the program will crash.

4.3. Managing Participants

The section below the properties is the Participants Section. Over the header you can view the options for several actions.

4.3.1. Managing via Excel

4.3.1.1. Uploading Excel List

Click the 'Add Participants via Excel' button. You will be met with a file explorer tab. Here you have to navigate to your desired excel file and select it to properly upload the list.

4.3.1.2. Exporting Excel List

Click the ‘Export Participant List via Excel’ button. You will be met with a file explorer tab. Navigate through your desired output directory and click ‘save’ when ready. The file will be exported to your selected directory.

4.3.2. Managing through the App

4.3.2.1. Add Participant

Scroll to the bottom of the list by using the scroll bar left to the participant list to find an empty row. Click the empty row, add relevant information for a participant. After doing so press ‘CTRL + S’ or click File=>Save to apply your changes.

4.3.2.2. Remove Participant

Select a row and press ‘DEL’. Save the file.

4.3.2.3. Clear Participant List

Click the ‘Clear Participant List’ button located over the participant list viewer menu.

4.3.2.4. Search Participant

Enter any property of a participant and search.

4.4. Generate Matchmakings

Navigate to the ‘Kumite’ tab using the tab viewer. Click the ‘Matchmake’ button located on the header of the tab.

4.5. Bracket Management

4.5.1.1. View Brackets

Select your desired age group over the top-left panel by clicking over the labels. Select the desired gender over the panel below the age group panel from the buttons beside the title labeled ‘Weight Category’. Select your desired weight category from the same panel. You may view the bracket below the header of the tab or to the left of the age group and weight category panels.

4.5.1.2. Modify Age Groups

Select your age group. Modify the age range using the section over the bottom of the panel. The left text box is the minimum age while the right text box is the maximum age. Enter the desired range by using the text boxes and press enter when satisfied.

The entered range must not overlap with the range of the other age groups or else the application won’t accept it.

4.5.1.2.1. *Add Age Groups*

Click the ‘Add Age Group’ button over the top-right corner of the Age Group panel. A new Age Group will be automatically created and selected. You have to modify the range for the changes to take effect. Don’t forget to press ‘enter’.

4.5.1.2.2. *Remove Age Groups*

Select an Age group. Press ‘Delete’.

4.5.1.3. Modify Weight Categories

Modify and select the weight categories same way as the Age Groups limited to the Remove and Range setting functions.

4.5.1.4. Generating/Adding Weight Categories

If you wish to generate or add weight categories first enter the minimum and maximum weights but do not press enter. After entering your desired interval and press enter this time. After you may click the button labeled 'Generate' and weight categories will be generated accordingly.

4.5.1.4.1. *Example*

If the minimum is 0, maximum is 100 and the interval is 20 the generated weight categories will be;

- 0-20
- 20-40
- 40-60
- 80-100
- 100+

The maximum weight is inclusive.

4.5.1.5. Exporting Bracket PDF

To export a pdf of your brackets, click the 'Export PDF' button located over the top-right corner of the bracket panel. Select your desired directory. Your pdf will be saved over there.

4.5.1.6. Swapping Participants

Click the label of a participant once, click the new desired position by clicking the other participant's label. The participants will be swapped with the matches numbered automatically. The number of the participants won't be changed if you wish to renumber them in accordance with the order of the matches you may refer to *4.6 Enumerize Elements.*

4.6. Enumerize Elements

To Numberize matches and participants you have to click the 'Numberize Elements' button over the bracket viewer. You may click on the right button to select either Numberize Matches or Numberize Participants. The program will only Numberize your selected elements. However, note that for the matches to be numbered their parent division has to be added to a proper Category Group. If the matches do not get numbered, please check if their division is added to a Category Group.

4.7. Category Group Management

For the matches to be numbered in respect to the groups that their divisions are a part of you may use Category groups.

4.7.1. Modifying Category Groups

4.7.1.1. Add Category Group

Click the Add Group button. A new group with an appropriate label will be created with no divisions.

4.7.1.2. Remove Category Group

Select a category group. Click the remove group button on the bottom.

4.7.2. Modify Added Divisions

Click the manage divisions button over the header panel when selecting a category group. Here you may select/deselect your desired divisions for your group. You may reorder the divisions by pressing and holding left mouse button whilst hovering a division label and dragging them to your desired position. The matches will be numbered in accordance with the position of your division within a category group.

4.7.3. View Matches List

You may view the list of the matches upon selecting a category group. The header shows information about the matches.

4.8. Match Management

You may manage matches of the tournament via several actions.

4.8.1. Qualifying a Participant/Declaring a Participant Winner

In the Brackets, you may double click a participant to declare them winner. You will see them advance to the next round.

Another option is to use the category group tab. Here you may click the label of the participant who won. By doing so the match will be flagged as complete thus it will turn blue whilst the next round will be updated both in the category group and the bracket view accordingly.